WORKED SOLUTION

STEP I 1992, Question 13

Mechanics • Escape Speed and Radial Fall

STEP I 1992, Question 13

Read the question before you go on. It is in the STEP Support Programme's question database:

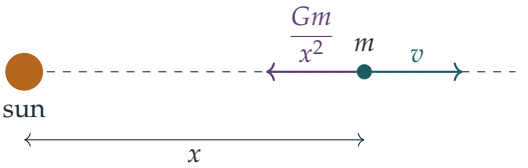
step.maths.org/questions/92-s1-q13

The key idea

The force depends only on the distance x, so the natural tool is the equation of motion written as $v\frac{\mathrm{d}v}{\mathrm{d}x}$, which integrates at once to an energy equation linking speed and distance. In the first part the fact that v^2 can never be negative turns that equation into a ceiling on x. In the second part the energy equation gives $\frac{\mathrm{d}x}{\mathrm{d}t}$ as a function of x, and separating the variables gives the time.

Solution

We model the sun as fixed and the comet as a particle, with no forces other than the given attraction. Since the attraction acts along the line joining the comet to the sun, a comet that is moving directly away from the sun (or is at rest) keeps moving along that line. So x is a coordinate along a straight line, and we let $v = \frac{\mathrm{d}x}{\mathrm{d}t}$, taking positive v to mean away from the sun.



The energy equation

Newton's second law along the line, with the force towards the sun (in the direction of decreasing x), gives

$$m\frac{\mathrm{d}^2x}{\mathrm{d}t^2} = -\frac{Gm}{x^2}.$$

Writing the acceleration as $v\frac{\mathrm{d}v}{\mathrm{d}x}$ and cancelling m,

$$v\frac{\mathrm{d}v}{\mathrm{d}x} = -\frac{G}{x^2} \qquad \Longrightarrow \qquad \tfrac{1}{2}v^2 = \frac{G}{x} + C$$

for some constant C. This holds throughout the motion, whichever way the comet is moving. (It is conservation of energy per unit mass: kinetic energy $\frac{1}{2}v^2$ plus potential energy $-G/x$.)

The condition for x to become arbitrarily large

At the given moment $x = h$ and $v = V$, so $C = \frac{1}{2}V^2 - \dfrac{G}{h}$ and, at every later time,

$$v^2 = V^2 - \frac{2G}{h} + \frac{2G}{x}.$$

We prove the contrapositive: if $V^2 < 2G/h$ then x is bounded.

Ask yourself The comet could in principle move about in all sorts of ways later on. How can one equation bound x for all time?

Because $v^2 \geq 0$ at every instant, whatever the comet is doing. Suppose $V^2 < 2G/h$, and put $k = \dfrac{2G}{h} - V^2 > 0$. Then the energy equation says

$$\frac{2G}{x} = v^2 + k \;\geq\; k, \qquad \text{so} \qquad x \leq \frac{2G}{k} = \frac{2Gh}{2G - hV^2}$$

at every later time (we may multiply up, since $x > 0$ and $k > 0$). So x can never exceed the fixed value $2Gh/(2G - hV^2)$: the comet rises to that distance, where $v = 0$, and then falls back. Hence x cannot become arbitrarily large unless $V^2 \geq 2G/h$, as required.

(The converse is also true. If $V^2 \geq 2G/h$ then $v^2 \geq 2G/x > 0$ while the comet moves outwards, so it never stops, and since $v^2 \geq 2G/x \geq 2G/X$ whenever $x \leq X$, it cannot linger below any distance X for ever. This is the escape speed.)

The time to fall into the sun

Now the comet starts at rest at a very great distance D. Released from rest, it falls straight towards the sun. By the energy equation with $v = 0$ at $x = D$,

$$\tfrac{1}{2}v^2 = \frac{G}{x} - \frac{G}{D}.$$

Ask yourself What does "a great distance" mean for the constant?

It means we let $D \to \infty$, so that $G/D \to 0$ and the constant is zero: the comet has zero total energy. (A large but finite D only makes the comet a little slower, and the time a little longer, by an amount that tends to zero as $D \to \infty$.) So at distance x its speed is $\sqrt{2G/x}$ and, since x is decreasing,

$$\frac{\mathrm{d}x}{\mathrm{d}t} = -\sqrt{\frac{2G}{x}} \qquad \Longrightarrow \qquad \sqrt{x}\,\mathrm{d}x = -\sqrt{2G}\,\mathrm{d}t.$$

Let T be the time to go from $x = h$ to $x = 0$. Integrating,

$$\sqrt{2G}\,T = \int_0^h \sqrt{x}\,\mathrm{d}x = \tfrac{2}{3}h^{3/2},$$

so

$$T = \frac{2h^{3/2}}{3\sqrt{2G}} = \frac{h}{3}\sqrt{\frac{2h}{G}}.$$

(The integrand $\sqrt{x}$ is perfectly well behaved at $x = 0$, so the comet really does reach the sun in finite time, even though its speed grows without limit.)

Checks

- Units: the force Gm/x^2 is a mass times an acceleration, so G has units $\mathrm{m^3\,s^{-2}}$. Then V^2 and $2G/h$ are both in $\mathrm{m^2\,s^{-2}}$, and $h^{3/2}/\sqrt{G}$ is in seconds.
- At $x = h$ the comet's speed is $\sqrt{2G/h}$. If it kept that speed it would take $h/\sqrt{2G/h} = \frac{h}{2}\sqrt{2h/G}$, which is longer than T, as it should be since the comet speeds up. In fact T is two-thirds of this.
- Numerically, with $G = h = 1$: stepping the motion forward from $x = 1$ with $v = \sqrt{1.5}$, the comet turns round at $x = 4.000$, as $2Gh/(2G - hV^2) = 4$ predicts. Starting from $x = 1$ with $v = -\sqrt{2}$, it reaches the sun after 0.4714, matching $T = \frac{1}{3}\sqrt{2} \approx 0.47140$. Starting from rest at $D = 10^6$ instead of infinity changes T only in the seventh decimal place.

Common slips

- Forgetting the minus sign in $\frac{\mathrm{d}x}{\mathrm{d}t} = -\sqrt{2G/x}$, then getting a negative time, or integrating from 0 to h without saying why.
- Showing only that the comet slows down: every comet moving away slows down, even one that escapes. The point is that, when $V^2 < 2G/h$, v^2 would become negative beyond a certain distance.

Beyond STEP

The quantity $\frac{1}{2}v^2 - G/x$ is the energy per unit mass in a Newtonian gravitational field, and its sign decides between bound orbits (ellipses) and escape (parabolas and hyperbolas). The zero-energy fall in the last part is the degenerate, straight-line case of a parabolic orbit, and $T \propto h^{3/2}$ is a relative of Kepler's third law. First-year courses on orbits develop all of this from the same energy and angular-momentum equations; Kleppner and Kolenkow, *An Introduction to Mechanics*, treats central forces and orbits thoroughly.

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